Mod Bus 1.01

General operation:

The Master (Aertecnica central power unit) communicates via broadcasts, using the user ID = 0 when sending. In order to speed up visualisations, the “write multiple register” mode is pre-set within Master, which sends all the data from the below described registers as one single string. Device managers are able to receive and view the data without sending any confirmation to the Master; in the instance where data is lost, the subsequent string is updated.

In the instance of problems receiving the entire string with “write multiple register”, the “write single register” mode is available in Master, which sends a register for every string transmitted, although, obviously, the speed governing transmission of information is lost.

The device managers have the option of resetting the Master card by sending a reset request. The aforementioned request is a “write single register” to a subsequently provided address. This string may be sent to the Master using the personal ID code “1”; in such a case, the Master responds as a standard slave (repeating the request in response). However it can also be sent to the Master using broadcast, in such a case, no response is provided. This request is sent from the device manager who received the reset request from the user via a dedicated input. The transmission takes place immediately after receipt of a valid string for the device manager, in compliance with the minimum latency period described below.

In the instance where the request is not received, it can be repeated immediately after the subsequent valid string up until the time it is correctly received. The Master only confirms receipt of the request if it was sent with the personal ID; in any case, it continues to send the standard string which features the Master card status, in such a way, the device manager can identify whether the reset request was received even where the request was sent via broadcast.

To simplify sending of the reset request, in the case of locks, the Master sends a broadcast string every 2 seconds, where normally it sends 4 per second. In the case of a lock owing to a high temperature, the reset is only permitted once a specific reset threshold value has been achieved; this status is sent in bit format as part of the Master status register.

The structure of the Bus system and reset request, in this way, prevents the installer from establishing too many settings, which would involve indicating how many device managers exist within the system, identifying them all separately and the Master carrying out polling operations. The described structure enables the simple connection of one or more device managers to ensure operation.

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Serial Line Data:

Communication protocol used:	MODBUS RTU
Serial data format:	8 Data Bits, 1 stop bit, No Parity, No Flow Control
Transfer speed:	19.200 Bps
Controls used:	Write Single Register, Write Multiple Register
Sending mode:	Broadcast (the Master communicates with everybody using the ID = 0 when sending)
Master ID (Aertecnica Card):	1
Display ID (device managers):	10 (all IDs can be used except ID 1)
Latency Period before received by Masterr	10 mS (the time in which the Master prepares for the receipt subse quent to the last transmission)(2 seconds are available, so a longer time period may be used, for example, 500 mS)

Sending data description:

The Master communicates via broadcasts, using the user ID = 0 when sending.
In order to speed up visualisations, the "write multiple register" mode is pre-set within Master, which sends all the data from the below described registers as one single string. Device managers are able to receive and view the data without sending any confirmation to the Master; in the instance where data is lost, the subsequent string is updated.

Registers used when sending data:

(4000)	01	0x0000	Unsigned Word = Card Model	from 0 to 65535
	0=Not identified			
	1=Perfect			
	2=Classic			

(4000) 02 0x0001	Unsigned Word = Interpretation of Register Content	16 Bit
Bit0=Card Hours	1=Without Decimal / 0=with Decimal	
Bit1= Motor Hours	1=Without Decimal / 0=with Decimal	
Bit2= Bag Hours	1=Without Decimal / 0=with Decimal	
Bit3= Filter Hours	1=Without Decimal / 0=with Decimal	
Bit4= Temperature	1=Without Decimal / 0=with Decimal	
Bit5= Maximum Time	1=Total Minutes / 0=Total Seconds	
Bit6=0		
Bit7=0		
Bit8=0		
Bit9=0		
Bit10=0		
Bit11=0		
Bit12=0		
Bit13=0		
Bit14=0		
Bit15=0		

(4000) 03 0x0002 Unsigned Word = Card Hours with decimal from 0 to 6553.5

(Where Register 2 features the relative greater Bit, the content should be interpreted without the decimal)

(4000) 04 0x0003 Unsigned Word = Motor Hours with decimal from 0 to 6553.5

(Where Register 2 features the relative greater Bit, the content should be interpreted without the decimal)

(4000) 05 0x0004 Unsigned Word = Bag Hours with decimal from 0 to 6553.5

(Where Register 2 features the relative greater Bit, the content should be interpreted without the decimal)

(4000) 06 0x0005 Unsigned Word = Filter Hours with decimal from 0 to 6553.5

(Where Register 2 features the relative greater Bit, the content should be interpreted without the decimal)

Registers used when sending data:

(4000) 07	0x0006	Unsigned Word = Bag Level	from 0 to 5	levels
(4000) 08	0x0007	Unsigned Word = Filter Level	from 0 to 5	Levels
(4000) 09	0x0008	Unsigned Word = Pressure Level	from 0 to 3	0= Off 1=Lo 2=ok 3=Hi
(4000) 10	0x0009	Unsigned Word = Pressure 1	from 0 to 500	mBar
(4000) 11	0x000A	Unsigned Word = Pressure 2	from 0 to 500	mBar
(4000) 12	0x000B	Unsigned Word = Press. Stat. Differential	from 0 to 500	mBar
(4000) 13	0x000C	Signed Word = Temperature	from -20.0 to 150.0 °C	
(4000) 14	0x000D	Unsigned Word = System Status	16 Bit Msb-Lsb	

Bit0+Bit1	=	PID Activation Status	
0	=	Disabled	
1	=	PID Maximum vacuum limiter	
2	=	PID maintenance of the pressure set point value	
Bit2	=	Micro Line Status	1= Closed / 0=Open
Bit3	=	Start Stop Status	
1	=	Start using the start stop button / 0 = Motor ON from the micro line or Motor OFF	
Bit4	=	Motor Status	1= ON / 0=OFF
Bit5	=	Disable aut. reset subsequent to an off motor	1 = Disabled / 0 = Enabled
Bit6	=	Stand By Status	1 = Active / 0 = Non Active
Bit7	=	Temperature Reset Control	1 = Prohibited / 0 = permitted
		(The temperature can only be reset where the temperature is below the threshold value, this bit indicates whether the temperature has fallen below this value)	

Note: Reset is only enabled where this bit is equal to Zero.

Bit8	=	Maximum Time Pre-alarm	1 = Active / 0 = Not active
Bit9	=	Maximum Pressure Pre-alarm	1 = Active / 0 = Not active
Bit10	=	Number of Starts Pre-alarm	1 = Active / 0 = Not active
Bit11	=	Dirty Filter Pre-alarm	1 = Active / 0 = Not active
Bit12	=	Full Bag Pre-alarm	1 = Active / 0 = Not active

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Registers used when sending data:

Bit13	=	Max Temperature Pre-alarm	1 = Active / 0 = Not active
Bit14	=	0	
Bit15	=	Appliance Pre-alarm	1 = Active / 0 = Not active

Note: This bit may be used to help identify that the central unit is in Pre-alarm status, this bit is always greater than the bit serving to identify the Fault

(4000) 15 0x000E Unsigned Word = Status of Locks 16 Bit Msb-Lsb

Bit0	=	Maximum Time Lock	1 = Active / 0 = Not active
Bit1	=	Maximum Pressure Lock	1 = Active / 0 = Not active
Bit2	=	Number of Starts Lock	1 = Active / 0 =Not active
Bit3	=	Dirty Filter Lock	1 = Active / 0 = Not active
Bit4	=	Full Bag Lock	1 = Active / 0 =Not active
Bit5	=	Max Temperature Lock	1 = Active / 0 = Not active
Bit6	=	0	
Bit7	=	Appliance Lock	1 = Active / 0 = Not active

Note: This bit may be used to help identify that the central power unit is in lock status, this bit is always greater than the bit serving to identify the Lock.

Bit8	=	0
Bit9	=	0
Bit10	=	0
Bit11	=	0
Bit12	=	0
Bit13	=	0
Bit14	=	0
Bit15	=	0

(4000) 16	0x000F	Unsigned Word = Motor Power Percentage	from 0 to 100%
(4000) 17	0x0010	Unsigned Word = Pressure Set Point	from 0 to 500 mBar
(4000) 18	0x0011	Unsigned Word = Residual maximum time	from 0 to 255 min./ from 0 to 15300 Sec

(Where Register 2 features the relative greater Bit, the content should be interpreted in minutes)

Description of the sending of the reset request:

The reset request may be sent as per two modes, via broadcasts or to a personal ID. In both cases, in compliance with the MODBUS protocol standard, with no response in the instance of broadcasts and with confirmation of receipt of the request in the case of a personal ID.

1) Broadcast Mode:

In this mode, the device manager who received the reset request from the user, sends the reset via broadcast with ID=0. The master who operates as a slave, does not confirm receipt, but upon termination of the latency period of the data to the device manager, sends a new string with the updated system status where receipt is confirmed, otherwise, it re-sends the data string which also includes the lock status.

2) Personal ID Mode (ID=1):

In this mode, the device manager receives the reset request from the user and sends the reset to the personal ID "1". The master who operates as a slave, sends the confirmation string in compliance with the MODBUS protocol standard, in particular repeating the request received, subsequently sending the data string which includes the lock status.

Register used to send the reset request and for the serial start/stop control

(4000)	64	0x003F	Unsigned Word = Reset Control	16 Bit Msb - Lsb
Bit0	=		Lock Reset Control	1 = Active / 0 = Not active
Bit1	=		Full Bag Reset Control	1 = Active / 0 = Not active
Bit2	=		Dirty Filter Reset Control	1 = Active / 0 =Not active
Bit3	=		Motor Hours Reset Control	1 = Active / 0 = Not active
Bit4	=		Serial Start Control	1 = Start / 0 =no action
Bit5	=		Serial Stop Control	1 = Start / 0=no action
Bit6	=	0		
Bit7	=	0		
Bit8	=	0		
Bit9	=	0		
Bit10	=	0		
Bit11	=	0		
Bit12	=	0	Bit14 = 0	
Bit13	=	0	Bit15 = 0	

the advanced vacuum system.

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